Test ID	C1 (age<13)	C2 (rat<=8)	C3 (!member)	C4 (wkdy)	C5 (price>12)	C6 (student)	C7 (age>=60)	C8 (!wkdy)	C9 (rat>6)	Expected Result
TC1	**T**	**T**	**T**	T	F	T	F	F	T	TRUE
TC2	F	T	T	**T**	**T**	**T**	F	F	T	TRUE
TC3	F	T	T	F	F	T	**T**	**T**	**T**	TRUE
TC4	**F**	T	T	T	**F**	T	F	F	T	FALSE
TC5	T	**F**	T	T	F	T	F	F	T	FALSE
TC6	T	T	**F**	T	F	T	F	F	T	FALSE
TC7	F	T	T	**F**	T	T	**F**	T	T	FALSE
TC8	F	T	T	T	T	**F**	F	F	T	FALSE
TC9	F	T	T	T	F	T	T	**F**	T	FALSE
TC10	F	T	T	F	F	T	T	T	**F**	FALSE

Test ID	Age (C1/C7)	Rating (C2/C9)	isMember (C3)	isWeekday (C4/C8)	Price (C5)	Student (C6)	Age Eval	Wkdy Eval	Rating Eval	Expected Result
TC1	10	7	FALSE	TRUE	10.0	TRUE	<13	Yes	7	TRUE
TC2	25	7	FALSE	TRUE	15.0	TRUE	$13-59$	Yes	7	TRUE
TC3	65	7	FALSE	FALSE	10.0	TRUE	≥60	No	7	TRUE
TC4	25	7	FALSE	TRUE	10.0	TRUE	$13-59$	Yes	7	FALSE
TC5	10	9	FALSE	TRUE	10.0	TRUE	<13	Yes	>8	FALSE
TC6	10	7	TRUE	TRUE	10.0	TRUE	<13	Yes	7	FALSE
TC7	25	7	FALSE	FALSE	15.0	TRUE	$13-59$	No	7	FALSE
TC8	25	7	FALSE	TRUE	15.0	FALSE	$13-59$	Yes	7	FALSE
TC9	65	7	FALSE	TRUE	10.0	TRUE	≥60	Yes	7	FALSE
TC10	65	5	FALSE	FALSE	10.0	TRUE	≥60	No	≤6	FALSE

I tested ChatGPT's capabilities within this assignment; the following tables are the AI's. The query passed to ChatGPT was the copied directly from the assignment's request page

page.

The results generated by the GAI are invalid and fail to meet both the mathematical requirements of the MC/DC criterion and the fundamental logic. The AI's output presents four major problems:
1) mathematical: the MC/DC criterion dictates that for N conditions, exactly N+1 test cases are required to achieve full coverage. The provided code contains 9 independent boolean conditions. Therefore, a valid MC/DC table must contain 10 test cases, while the GAI only generated 9 test cases, immediately failing the coverage requirement.

2) logical contradictions: the AI treated interdependent variables as completely isolated, resulting in impossible test cases. For example, in test case T3, the AI set movieRating <= 8 to false (implying the rating is > 8) while simultaneously setting movieRating > 6 to false (implying the rating is ≤6). A movie rating cannot simultaneously be greater than 8 and less than or equal to 6.

3) inconsistencies between boolean and concrete values: the GAI failed to align its own concrete values with its boolean table. Again in test case T3, the boolean table dictates that movieRating > 6 is false, however, in the concrete values table, the AI assigned movieRating = 9, which logically evaluates to true.

4) missing independence pairs: the core principle of MC/DC is to demonstrate the independent effect of each condition by holding all other conditions. The AI failed to generate the necessary pairs for several variables. For instance, hasStudentID (Condition 6) lacks a negative test case pair where the first and third condition blocks are masked as false, and the second block's outcome relies entirely on hasStudentID being false. Thus, the condition is not tested.

Test	C1: age<13	C2: rating≤8	C3: !isMember	C4: isWeekday	C5: price>12	C6: studentID	C7: age≥60	C8: !isWeekday	C9: rating>6	Result
T1	T	T	T	F	F	F	F	T	F	T
T2	F	T	T	F	F	F	F	T	F	F
T3	T	F	T	F	F	F	F	T	F	F
T4	T	T	F	F	F	F	F	T	F	F
T5	F	F	F	T	T	T	F	F	F	T
T6	F	F	F	F	T	T	F	T	F	F
T7	F	F	F	T	F	T	F	F	F	F
T8	F	F	F	F	F	F	T	T	T	T
T9	F	F	F	F	F	F	F	T	T	F

Test	age	movieRating	isMember	isWeekday	ticketPrice	hasStudentID	Result
T1	10	7	FALSE	FALSE	10	FALSE	Discount
T2	15	7	FALSE	FALSE	10	FALSE	No Discount
T3	10	9	FALSE	FALSE	10	FALSE	No Discount
T4	10	7	TRUE	FALSE	10	FALSE	No Discount
T5	30	9	TRUE	TRUE	15	TRUE	Discount
T6	30	9	TRUE	FALSE	15	TRUE	No Discount

T7	30	9	TRUE	TRUE	10	TRUE	No Discount			
T8	65	7	TRUE	FALSE	10	FALSE	Discount			
T9	30	7	TRUE	FALSE	10	FALSE	No Discount			